

KARTHIKA GANESAN

Career Objective

Motivated Electronics and Communication Engineering graduate seeking an opportunity at Kaar Technologies to utilize my knowledge in electronics, embedded systems, and programming along with strong logical thinking and problem-solving skills, while learning enterprise software technologies and contributing to efficient and innovative solutions.



Education

- **B.Tech – Electronics and Communication Engineering**
Sri Manakula Vinayagar Engineering College, Puducherry (2023 – 2027)
Currently pursuing.
CGPA: 8.78 (till Semester V)
- **Higher Secondary Education** (2022 - 2023)
Amalorpavam Higher Secondary School, Puducherry
Percentage: 86.16%
- **High School** (2020 - 2021)
Amalorpavam Higher Secondary School, Puducherry
Percentage: 80%

Projects

- **Smart Pressure Monitoring Bed – Embedded Systems Project**
Developed an IoT-based system to perform pressure distribution using sensors to prevent pressure ulcers.
- **IoT-Based Cycle Counter For Packaging Machines With Maintenance Alerts**
Developed an IoT system with sensors and microcontroller to track machine cycles and generate maintenance alerts.
- **Smart Helmet for Rider Safety (Work in Progress)**
Designing an IoT-based smart helmet aimed at detecting accidents and monitoring rider safety.
Planning to integrate features such as rule-violation detection and automated alert system.
Currently working on hardware integration and data transmission setup.

Hobbies & Interests

- Painting and Art works
- Learning new technologies and software tools
- Listening to music and podcasts

Skills

- **Programming:** C, Java(Basics), Verilog(Basics)
- **Core Subjects:** Digital Electronics, Microcontrollers, Signals & Systems, Communication Systems
- **Tools:** Arduino IDE, Vivado, Cisco Packet Tracer
- **Hardware:** Arduino, Sensors, Raspberry Pi R2040, ESP8266

Certifications

- **Introduction to IoT – NPTEL** (Elite silver – 85%)
- **CCNA: Switching, Routing, and Wireless Essentials – Cisco** Networking Academy
- **ARM® Cortex®-M Architecture Overview – Microchip** University
- **Programming in Java – Ethnotech**
- **Web development - Ethnotech**
- **VLSI Design and Verification – Taras** System and Solutions

Internship

Embedded system Internship – VI microsystems, Chennai.

- Gained practical exposure to embedded systems design and development.
- Developed basic understanding of embedded C programming and system architecture.

Soft Skills

- Problem-solving ability
- Logical and analytical thinking
- Teamwork and collaboration
- Adaptability and quick learning

Technical Activities

Technical Paper Presentation – Velammal Institute of Technology

Topic: IoT-Based Flour Grinder Cleaning System

- Presented a paper on an IoT-based automation system for cleaning flour grinding machines.
- Explained the use of sensors and microcontrollers for remote monitoring and control.

Hackathon Participant – Jeppiaar Institute of Technology

Project: Women Safety Device with Defense Mechanism

- Participated in a hackathon and proposed a women safety device using embedded systems.
- Designed a system capable of triggering emergency alerts during unsafe situations.
- Collaborated in a team to develop innovative safety solutions.

Ideathon – Puducherry Technological university

Topic: YOLO-Based CCTV Theft Detection System (Concept Presentation)

- Presented the concept of a theft detection system using CCTV surveillance and YOLO object detection.
- Proposed a method to identify suspicious activities and unauthorized object removal from video footage.

Extra-Curricular Activities

- Active member of Institution of Engineers (India) (IEI) student chapter and Indian Society for Technical Education (ISTE) club.
- Delivered a seminar on **Electrical Technology Concept – Servo Motor**.
- Organized and coordinated **technical events such as quizzes and student competitions**.
- Participated in **technical discussions and knowledge-sharing activities** through engineering clubs.

Industrial Exposure

- Industrial visit to Techvolt Software Pvt Ltd to understand industry practices.
- Attended **Automation ExpoSouth (2nd Regional Automation Expo in Chennai)** to explore recent technologies in automation and industrial systems.
- Strong interest in gaining practical experience through real-world applications.
- Ability to analyze situations logically and approach problems with structured thinking.